

Scientific Communities in the Literature Selection

Qualitative category coding based exclusively on the fields **title** and **keywords** in the uploaded CSV file.

Dataset: 250 records. Coding approach: inductive thematic aggregation supported by transparent keyword/title matching. Categories are not mutually exclusive; one article can represent multiple communities when title/keywords point to several fields.

Overview of coded communities

Community	N	Definition	Example from the dataset
Assessment, Measurement and Psychometrics	81	Communities focused on tests, scales, rubrics, measurement models, validation, grading, student evaluations, and psychometric methodology.	<i>Development and validation of a rapid and precise online sentence reading efficiency assessment</i> Keywords: dyslexia; reading fluency assessment; screening tools; reading fluency and comprehension; progress monitoring; psychometrics
Educational Technology, AI and Learning Analytics	65	Work on digital tools, artificial intelligence, online platforms, learning analytics, NLP, machine learning, data science, and technology-mediated learning environments.	<i>Content Analysis of Textbooks via Natural Language Processing: Findings on Gender, Race, and Ethnicity in Texas U.S. History Textbooks</i> Keywords: artificial intelligence; case studies; content analysis; curriculum; data science; gender studies; history; natural language processing; race; textbooks; textual analysis
Motivation, Emotion, Well-being and Educational Psychology	59	Educational psychology communities studying motivation, emotion, self-efficacy, self-regulation, identity, personality, stress, mental health, well-being, attitudes, and engagement.	<i>Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources</i> Keywords: COVID-19 pandemic, college students, psychological well-being, emotions, experience sampling method, personality, intraindividual trajectories
Higher Education and Student Success	51	Studies situated in colleges and universities, including undergraduate education, graduate education, doctoral training, access, retention, belonging, advising, and postsecondary outcomes.	<i>Going to College With a Posse: How Having High School Peers on Campus Supports College Achievement</i> Keywords: admissions; equity; first-generation college students; hierarchical linear modeling; higher education; peer interaction/friendship; privilege; psychology; regression analyses; retention; social belonging; social capital; socioeconomic status
Childhood, Schooling and Developmental Contexts	47	Research centered on children, adolescents, primary and secondary schooling, developmental trajectories, classroom contexts, and school-level learning environments.	<i>Determine the impact of Emotive Intelligent Spaces on childrens behavioural and cognitive outcomes</i> Keywords: early childhood; colour; intervention; working memory; self-regulation
Teacher Education and Professional Learning	46	Research on teachers, teacher candidates, professional development, professional learning networks, teacher beliefs, teacher identity, and classroom practice.	<i>A Different Experience in a Different Moment? Teachers Social Media Use Before and During the COVID-19 Pandemic</i> Keywords: online spaces; teacher learning; professional learning networks; COVID-19; social media; stress
STEM, Science and Mathematics Education	41	Research communities concerned with disciplinary teaching and learning in science, technology, engineering, mathematics, biology, chemistry, physics, and related STEM domains.	<i>Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database</i> Keywords: ICT; mathematics education; bibliometric analysis; Scopus

Community	N	Definition	Example from the dataset
Equity, Inclusion and Diversity in Education	39	Work addressing inclusive education, special education, disability, socioeconomic status, gender, race, ethnicity, equity, diversity, social justice, and access inequalities.	<i>Towards Quality Education for All: Integrating EdTech, Mentorship, and Community in Support of SDG 4</i> Keywords: EdTech; inclusive education; STEM education; mentorship; service learning; digital tools; remote learning; educational equity
Reading, Language and Literacy Education	31	Communities focused on reading, writing, literacy, language learning, bilingualism, fluency, comprehension, vocabulary, and linguistic development.	<i>Development and validation of a rapid and precise online sentence reading efficiency assessment</i> Keywords: dyslexia; reading fluency assessment; screening tools; reading fluency and comprehension; progress monitoring; psychometrics
Online, Distance and Pandemic-Related Education	25	Communities examining online education, distance learning, blended or hybrid learning, emergency remote teaching, and COVID-19-related educational change.	<i>Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources</i> Keywords: COVID-19 pandemic, college students, psychological well-being, emotions, experience sampling method, personality, intraindividual trajectories
Immersive, Game-Based and Simulation Learning	25	Communities studying virtual reality, augmented reality, serious games, game-based learning, simulations, and immersive technologies for education.	<i>The more the better? Comparing two SQD-based learning designs in a teacher training on augmented and virtual reality</i> Keywords: WillSkillTool model; Teacher education; Teacher training; SQD model; Augmented reality; Virtual reality; Learning design; Learning events
Open Science, Research Infrastructure and Meta-Research	22	Meta-scientific work on open science practices, open data/code/materials, preregistration, bibliometrics, systematic reviews, meta-analyses, research transparency, and methodological infrastructure.	<i>Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database</i> Keywords: ICT; mathematics education; bibliometric analysis; Scopus
Educational Policy, Reform and Public Discourse	16	Studies on educational reforms, standards, public sentiment, policy thresholds, accountability, communication about science or education, and public discourse around education.	<i>Understanding Public Sentiment About Educational Reforms: The Next Generation Science Standards on Twitter</i> Keywords: public sentiment, Next Generation Science Standards, sentiment analysis, educational data science
Curriculum, Textbooks and Educational Content	8	Studies of curricula, textbooks, learning materials, content analysis, course design, instructional resources, and representation of knowledge in educational materials.	<i>Content Analysis of Textbooks via Natural Language Processing: Findings on Gender, Race, and Ethnicity in Texas U.S. History Textbooks</i> Keywords: artificial intelligence; case studies; content analysis; curriculum; data science; gender studies; history; natural language processing; race; textbooks; textual analysis

Interpretive notes

The strongest visible communities are not single disciplines only; they combine subject domains, methodological infrastructures, and educational settings. The largest clusters point to assessment and psychometrics, STEM/science education, higher education, educational technology/AI, and educational psychology. This reflects a literature selection in which education research is strongly connected to measurement, data-intensive methods, higher-education outcomes, and digitally mediated learning.

Because only titles and keywords were used, the labels should be understood as thematic community indicators rather than definitive classifications of full papers. Counts indicate how often a community signal appears in titles/keywords; they do not sum to 250 because the coding is multi-label.

Category definitions and examples

Assessment, Measurement and Psychometrics (n = 81)

Communities focused on tests, scales, rubrics, measurement models, validation, grading, student evaluations, and psychometric methodology.

Example: *Development and validation of a rapid and precise online sentence reading efficiency assessment*

Example keywords: dyslexia; reading fluency assessment; screening tools; reading fluency and comprehension; progress monitoring; psychometrics

Educational Technology, AI and Learning Analytics (n = 65)

Work on digital tools, artificial intelligence, online platforms, learning analytics, NLP, machine learning, data science, and technology-mediated learning environments.

Example: *Content Analysis of Textbooks via Natural Language Processing: Findings on Gender, Race, and Ethnicity in Texas U.S. History Textbooks*

Example keywords: artificial intelligence; case studies; content analysis; curriculum; data science; gender studies; history; natural language processing; race; textbooks; textual analysis

Motivation, Emotion, Well-being and Educational Psychology (n = 59)

Educational psychology communities studying motivation, emotion, self-efficacy, self-regulation, identity, personality, stress, mental health, well-being, attitudes, and engagement.

Example: *Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources*

Example keywords: COVID-19 pandemic, college students, psychological well-being, emotions, experience sampling method, personality, intraindividual trajectories

Higher Education and Student Success (n = 51)

Studies situated in colleges and universities, including undergraduate education, graduate education, doctoral training, access, retention, belonging, advising, and postsecondary outcomes.

Example: *Going to College With a Posse: How Having High School Peers on Campus Supports College Achievement*

Example keywords: admissions; equity; first-generation college students; hierarchical linear modeling; higher education; peer interaction/friendship; privilege; psychology; regression analyses; retention; social belonging; social capital; socioeconomic status

Childhood, Schooling and Developmental Contexts (n = 47)

Research centered on children, adolescents, primary and secondary schooling, developmental trajectories, classroom contexts, and school-level learning environments.

Example: *Determine the impact of Emotive Intelligent Spaces on childrens behavioural and cognitive outcomes*

Example keywords: early childhood; colour; intervention; working memory; self-regulation

Teacher Education and Professional Learning (n = 46)

Research on teachers, teacher candidates, professional development, professional learning networks, teacher beliefs, teacher identity, and classroom practice.

Example: *A Different Experience in a Different Moment? Teachers Social Media Use Before and During the COVID-19 Pandemic*

Example keywords: online spaces; teacher learning; professional learning networks; COVID-19; social media; stress

STEM, Science and Mathematics Education (n = 41)

Research communities concerned with disciplinary teaching and learning in science, technology, engineering, mathematics, biology, chemistry, physics, and related STEM domains.

Example: *Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database*

Example keywords: ICT; mathematics education; bibliometric analysis; Scopus

Equity, Inclusion and Diversity in Education (n = 39)

Work addressing inclusive education, special education, disability, socioeconomic status, gender, race, ethnicity, equity, diversity, social justice, and access inequalities.

Example: *Towards Quality Education for All: Integrating EdTech, Mentorship, and Community in Support of SDG 4*

Example keywords: EdTech; inclusive education; STEM education; mentorship; service learning; digital tools; remote learning; educational equity

Reading, Language and Literacy Education (n = 31)

Communities focused on reading, writing, literacy, language learning, bilingualism, fluency, comprehension, vocabulary, and linguistic development.

Example: *Development and validation of a rapid and precise online sentence reading efficiency assessment*

Example keywords: dyslexia; reading fluency assessment; screening tools; reading fluency and comprehension; progress monitoring; psychometrics

Online, Distance and Pandemic-Related Education (n = 25)

Communities examining online education, distance learning, blended or hybrid learning, emergency remote teaching, and COVID-19-related educational change.

Example: *Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources*

Example keywords: COVID-19 pandemic, college students, psychological well-being, emotions, experience sampling method, personality, intraindividual trajectories

Immersive, Game-Based and Simulation Learning (n = 25)

Communities studying virtual reality, augmented reality, serious games, game-based learning, simulations, and immersive technologies for education.

Example: *The more the better? Comparing two SQD-based learning designs in a teacher training on augmented and virtual reality*

Example keywords: WillSkillTool model; Teacher education; Teacher training; SQD model; Augmented reality; Virtual reality; Learning design; Learning events

Open Science, Research Infrastructure and Meta-Research (n = 22)

Meta-scientific work on open science practices, open data/code/materials, preregistration, bibliometrics, systematic reviews, meta-analyses, research transparency, and methodological infrastructure.

Example: *Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database*

Example keywords: ICT; mathematics education; bibliometric analysis; Scopus

Educational Policy, Reform and Public Discourse (n = 16)

Studies on educational reforms, standards, public sentiment, policy thresholds, accountability, communication about science or education, and public discourse around education.

Example: *Understanding Public Sentiment About Educational Reforms: The Next Generation Science Standards on Twitter*

Example keywords: public sentiment, Next Generation Science Standards, sentiment analysis, educational data science

Curriculum, Textbooks and Educational Content (n = 8)

Studies of curricula, textbooks, learning materials, content analysis, course design, instructional resources, and representation of knowledge in educational materials.

Example: *Content Analysis of Textbooks via Natural Language Processing: Findings on Gender, Race, and Ethnicity in Texas U.S. History Textbooks*

Example keywords: artificial intelligence; case studies; content analysis; curriculum; data science; gender studies; history; natural language processing; race; textbooks; textual analysis

Coding rules used for transparency

Community	Typical title/keyword signals
STEM, Science and Mathematics Education	stem, science education, mathematics, biology, chemistry, physics, engineering education, bioinformatics
Educational Technology, AI and Learning Analytics	artificial intelligence, ai, generative ai, chatgpt, machine learning, natural language processing, learning analytics, educational data science
Higher Education and Student Success	higher education, college, university, undergraduate, graduate education, doctoral, phd, postsecondary
Assessment, Measurement and Psychometrics	assessment, psychometric, measurement, validation, scale, test, evaluation, rubric
Teacher Education and Professional Learning	teacher education, professional development, professional learning, pre-service, preservice, in-service, inservice, teacher
Equity, Inclusion and Diversity in Education	inclusive education, special education, disability, equity, diversity, gender, race, ethnicity
Motivation, Emotion, Well-being and Educational Psychology	motivation, emotion, well-being, wellbeing, mental health, stress, self-efficacy, self-regulation
Reading, Language and Literacy Education	reading, literacy, language learning, writing, bilingual, fluency, comprehension, vocabulary
Curriculum, Textbooks and Educational Content	curriculum, textbook, textbooks, learning materials, instructional materials, course design, content analysis, educational content
Online, Distance and Pandemic-Related Education	covid, pandemic, online education, online learning, distance learning, remote learning, remote teaching, blended learning
Open Science, Research Infrastructure and Meta-Research	open science, open data, open code, open materials, preregistration, pre-registration, bibliometric, systematic review
Immersive, Game-Based and Simulation Learning	virtual reality, augmented reality, vr, ar, serious games, game-based, gamification, simulation
Educational Policy, Reform and Public Discourse	policy, reform, public sentiment, standards, accountability, attendance policy, absenteeism, science communication
Childhood, Schooling and Developmental Contexts	children, adolescents, primary education, secondary education, high school, elementary, kindergarten, early childhood

Uncoded records: 13

These records did not contain strong title/keyword signals for the defined communities and may need manual review if exhaustive classification is required. Examples: An Interactive Web Application Helps Students Explore Water Balance Concepts; A Brief Parent-Coaching Package for Tiered Language Interventions; Disadvantaged by chance? Examining the persistence of relative age effects on educational achievement; Effects of Self-Explaining on Learning and Transfer of Critical Thinking Skills; Tone superimposition technique in Speech Sciences: a tutorial